Shitao Jin

AI in Architecture | Collective Intelligence | Megaproject Decision-Making

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24 published papers (20 journal articles + 4 conference papers) • 16 first- or corresponding-author peer-reviewed journal articles • 320 Google Scholar citations • h-index / i10-index: 8 / 7 • 183 verified peer reviews

Metrics verified from Google Scholar, Web of Science, and ORCID, August 2026.

ACADEMIC PROFILE

PhD candidate in Architecture at Tongji University and visiting PhD researcher at The University of Melbourne. My research develops AI-enabled architectural programming and collective-intelligence decision systems for complex built-environment problems, especially megaproject governance, stakeholder coordination, risk and complexity assessment, public concern analysis, and value-based evaluation of human-centered environments. I combine architectural knowledge with multi-agent reinforcement learning, natural language processing, bibliometrics, social network analysis, fuzzy-grey decision methods, and empirical design evaluation.

RESEARCH CLUSTERS AND PI POTENTIAL

1. AI-enabled architectural programming and collective-intelligence decision-making.

- Treats architectural programming as a front-end decision system for problem definition, value clarification, stakeholder alignment, and evidence-based evaluation.
- Develops group decision-making and multi-agent reinforcement learning frameworks for mega transportation projects and public infrastructure programming.
- Representative outputs: ECAM 2024 group decision-making model; CAADRIA 2026 multi-agent reinforcement learning paper; AEDM 2025 review of collective intelligence in construction.
- Grant-ready direction: AI-supported briefing platforms that integrate expert judgement, public needs, and adaptive optimization for early-stage public project decisions.

2. Megaproject governance, complexity, and stakeholder coordination.

- Builds computational evidence for megaproject risk, complexity, stakeholder networks, and dynamic public concerns.
- Integrates SNA, MCDM, gray fuzzy group decision-making, Monte Carlo simulation, topic modeling, sentiment analysis, and systematic review.
- Representative outputs: JCEM 2026 public concern dynamics; JAE 2026 risk assessment; ECAM 2026 complexity measurement; ECAM 2025 stakeholder identification.
- Grant-ready direction: data-driven governance toolkits for resilient, transparent, and stakeholder-responsive infrastructure delivery.

3. Human-centered and value-based built-environment evaluation.

- Extends architectural programming from buildings to communities, streets, learning environments, and child-friendly public spaces.
- Uses street-view imagery, questionnaire data, public reviews, AI perception models, and post-occupancy evaluation to translate human well-being into design evidence.
- Representative outputs: FoAR 2026 value-based programming for children's well-being; FoAR 2026 neighborhood center public perception analysis; Frontiers in Psychology 2022 classroom perception paper.
- Grant-ready direction: human-centered AI for community renewal, social infrastructure, and learning-space evaluation.

EDUCATION AND VISITING RESEARCH

Tongji University, Shanghai, China 2023 - 2027 expected

PhD in Architecture, Department of Architecture, College of Architecture and Urban Planning.

- Dissertation: Collective-Intelligence-Based Collaborative Decision-Making Systems for Architectural Programming.
- Research focus: architectural programming, AI-enabled group decision-making, collective intelligence, and megaproject management.
- GPA: 3.60/4.00. Supervisor: Prof. Huijun Tu.

The University of Melbourne, Melbourne, Australia 2025 - 2026

Visiting PhD Researcher, Department of Infrastructure Engineering, Faculty of Engineering and Information Technology.

- Research focus: value management, stakeholder coordination, and complexity governance in megaprojects.
- Joint-training advisors: A/Prof Felix Kin Peng Hui and A/Prof Paulo Vaz Serra.

Huazhong University of Science and Technology, Wuhan, China 2020 - 2023

Master of Architecture, Department of Architecture.

- Thesis: Design and Evaluation of Active-Learning Built Environments in Higher Education.
- GPA: 3.50/4.00. Supervisor: Prof. Lei Peng.

Jilin Jianzhu University, Changchun, China 2015 - 2020

Bachelor of Architecture, School of Architecture and Planning.

- GPA: 3.46/4.00.

SELECTED PUBLICATIONS (* denotes corresponding author; citations from Google Scholar, August 2026)

Shitao Jin*, "Dynamics and Influences Analysis of Public Concerns in Mega Construction Projects: An Integrated Topic and Sentiment Modeling Approach," *Journal of Construction Engineering and Management*, 2026, 152(8), 04026104. ASCE; SJR Q1.

Shitao Jin* and Xudong Miao, "Risk Identification and Assessment of Mega Construction Projects in Uncertain Environments: A Gray Fuzzy Group Decision-Making and Monte Carlo Simulation Method," *Journal of Architectural Engineering*, 2026, 32(2), 04026010. **[1 cite]** ASCE.

Huijun Tu, **Shitao Jin***, Xudong Miao, Cong Zhou, and Yuming Shi, "A Value-Based Programming Framework for Enhancing Children's Well-Being in Urban Communities," *Frontiers of Architectural Research*, 2026. SJR Q1.

Shitao Jin*, "Current Status and Trends of Megaproject Research: Bibliometric and Text Mining Analysis," *Engineering, Construction and Architectural Management*, 2026, 33(6), 4543-4573. **[8 cites]** 2024 IF 3.9; CiteScore 10.7.

Shitao Jin*, "Measuring Complexity in Mega Construction Projects: Fuzzy Comprehensive Evaluation and Grey Relational Analysis," *Engineering, Construction and Architectural Management*, 2026, 33(1), 284-317. **[7 cites]**

Shitao Jin*, "Interdisciplinary Perspective on Architectural Programming: Current Status and Future Directions," *Engineering, Construction and Architectural Management*, 2025, 32(11), 7332-7372. **[17 cites]**

Shitao Jin* and Yuan Deng, "Identification and Evaluation of Key Stakeholders in Mega Construction Projects: Social Network Analysis and Multi-Criteria Decision-Making," *Engineering, Construction and Architectural Management*, 2025, 1-38. **[1 cite]**

Shitao Jin*, "The Application of Collective Intelligence in the Construction Industry: A Review of the Current State, Challenges, and Opportunities," *Architectural Engineering and Design Management*, 2025, 21(5), 1091-1116. **[5 cites]**

Shitao Jin, Huijun Tu*, Jiangfeng Li, Yuwei Fang, Zhang Qu, Fan Xu, Kun Liu, and Yiqun Lin, "Enhancing Architectural Education Through Artificial Intelligence: A Case Study of an AI-Assisted Architectural Programming and Design Course," *Buildings*, 2024, 14(6), 1613. **[89 cites]** 2026 IF 3.4; JCR Q2.

Shitao Jin and Lei Peng, "Classroom Perception in Higher Education: The Impact of Spatial Factors on Student Satisfaction in Lecture Versus Active Learning Classrooms," *Frontiers in Psychology*, 2022, 13, 941285. **[56 cites]**

COMPLETE PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Shitao Jin***. Dynamics and Influences Analysis of Public Concerns in Mega Construction Projects: An Integrated Topic and Sentiment Modeling Approach. *Journal of Construction Engineering and Management*, 2026, 152(8), 04026104.

2. **Shitao Jin*** and Xudong Miao. Risk Identification and Assessment of Mega Construction Projects in Uncertain Environments: A Gray Fuzzy Group Decision-Making and Monte Carlo Simulation Method. *Journal of Architectural Engineering*, 2026, 32(2), 04026010. **[1 cite]**

3. Huijun Tu, **Shitao Jin***, Xudong Miao, Cong Zhou, and Yuming Shi. A Value-Based Programming Framework for Enhancing Children's Well-Being in Urban Communities. *Frontiers of Architectural Research*, 2026.

4. Yuan Deng, **Shitao Jin**, and Xiaoming Zhu. Evolving Public Perceptions of the Renewal of Suzhou Neighborhood Centers: An Integrated Sentiment and Topic Modeling Framework. *Frontiers of Architectural Research*, 2026.

5. Huijun Tu, Xudong Miao, and **Shitao Jin**. A Deep Learning-Based Evaluation System for Child-Friendly Urban Streets Integrating Abstract and Concrete Features: A Case of Shanghai Urban Street. *PLOS One*, 2026, 21(2), e0342430.

6. **Shitao Jin***. Current Status and Trends of Megaproject Research: Bibliometric and Text Mining Analysis. *Engineering, Construction and Architectural Management*, 2026, 33(6), 4543-4573. **[8 cites]**

7. **Shitao Jin***. Measuring Complexity in Mega Construction Projects: Fuzzy Comprehensive Evaluation and Grey Relational Analysis. *Engineering, Construction and Architectural Management*, 2026, 33(1), 284-317. **[7 cites]**

8. **Shitao Jin*** and Yuan Deng. Identification and Evaluation of Key Stakeholders in Mega Construction Projects: Social Network Analysis and Multi-Criteria Decision-Making. *Engineering, Construction and Architectural Management*, 2025, 1-38. **[1 cite]**

9. **Shitao Jin***. Interdisciplinary Perspective on Architectural Programming: Current Status and Future Directions. *Engineering, Construction and Architectural Management*, 2025, 32(11), 7332-7372. **[17 cites]**

10. **Shitao Jin***. The Application of Collective Intelligence in the Construction Industry: A Review of the Current State, Challenges, and Opportunities. *Architectural Engineering and Design Management*, 2025, 21(5), 1091-1116. **[5 cites]**

11. **Shitao Jin** and Huijun Tu*. Current Status and Research Progress in Architectural Programming: A Comparative Analysis Between China and Other Countries. *Frontiers of Architectural Research*, 2025, 14(2), 487-509. **[6 cites]**

12. Huijun Tu, Xudong Miao, **Shitao Jin**, Jiayi Yang, Xinyue Miao, and Jiale Qi. Deep Learning-Based Systems for Evaluating and Enhancing Child-Friendliness of Urban Streets: A Case of Shanghai Urban Street. *Buildings*, 2025, 15(13), 2291. **[7 cites]**

13. Huijun Tu, **Shitao Jin**, Tianlin Zhang, Jiangfeng Li, and Cong Zhou. Exploration of the Applications of Next-Generation Artificial Intelligence in Architectural Creation. *Architecture and Culture*, 2025(11), 289-291.

14. **Shitao Jin**, Huijun Tu*, Jiangfeng Li, Yuwei Fang, Zhang Qu, Fan Xu, Kun Liu, and Yiqun Lin. Enhancing Architectural Education Through Artificial Intelligence: A Case Study of an AI-Assisted Architectural Programming and Design Course. *Buildings*, 2024, 14(6), 1613. **[89 cites]**

15. Huijun Tu and **Shitao Jin***. A Group Decision-Making Model for Architectural Programming in Megaprojects. *Engineering, Construction and Architectural Management*, 2024, 31(13), 342-368. **[13 cites]**

16. **Shitao Jin** and Lei Peng. Classroom Perception in Higher Education: The Impact of Spatial Factors on Student Satisfaction in Lecture Versus Active Learning Classrooms. *Frontiers in Psychology*, 2022, 13, 941285. **[56 cites]**

17. Lei Peng, **Shitao Jin**, Yuan Deng, and Yichen Gong. Students' Perceptions of Active Learning Classrooms from an Informal Learning Perspective: Building a Full-Time Sustainable Learning Environment in Higher Education. *Sustainability*, 2022, 14(14), 8578. **[38 cites]**

18. Lei Peng, Wenli Wei, Wenyi Fan, **Shitao Jin**, and Yuxuan Liu. Student Experience and Satisfaction in Academic Libraries: A Comparative Study Among Three Universities in Wuhan. *Buildings*, 2022, 12(5), 682. **[30 cites]**

19. Lei Peng, Yuan Deng, and **Shitao Jin**. The Evaluation of Active Learning Classrooms: Impact of Spatial Factors on Students' Learning Experience and Learning Engagement. *Sustainability*, 2022, 14(8), 4839. **[40 cites]**

20. Lei Peng, Wenyi Fan, and **Shitao Jin**. Analysis of Campus Learning Environment Elements for Promoting Innovative Talent Cultivation: A Case Study of Campus Space Planning in Two Innovative Research Universities. *New Architecture*, 2022(6), 98-103. Chinese core journal.

Published Conference Papers

1. **Jin, S.***, Deng, Y., & Tu, H. (2026). Enabling intelligent collaboration in architectural programming: A hierarchical multi-agent reinforcement learning framework for mega transportation projects. In *Construction Research Congress 2026* (pp. 609-619). American Society of Civil Engineers.

2. Deng, Y., & **Jin, S.*** (2026). Collective intelligence for architectural programming in mega transportation projects: Multi-agent simulation and optimization. In *Construction Research Congress 2026* (pp. 675-685). American Society of Civil Engineers.

3. **Shitao Jin***, Yuan Deng, and Huijun Tu. Collective Intelligence in Architectural Programming: Multi-Agent Reinforcement Learning for Optimization in Mega Transportation Projects. *Proceedings of the 31st International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)*, 2026, Vol. 1, 693-702.

4. **Shitao Jin***, Yuan Deng, and Huijun Tu. Reprogramming Urban Communities: Enhancing Children's Well-Being Through a Value-Based Programming Approach in Inclusive Planning and Design. *Proceedings of the 60th ISOCARP World Planning Congress*, 2024, Siena, Italy. **[2 cites]**

MANUSCRIPTS AND RESEARCH IN PROGRESS

- Dissertation monograph: *Collective-Intelligence-Based Collaborative Decision-Making Systems for Architectural Programming*, expected 2027.
- Journal manuscript in development: multi-agent reinforcement learning for collaborative architectural programming in mega transportation hubs.
- Method paper in development: integrating topic modeling, sentiment analysis, and stakeholder networks for adaptive megaproject governance.
- Teaching research in development: scalable AI-assisted architectural programming pedagogy for problem seeking, brief generation, and design evaluation.

RESEARCH PROJECTS AND FUNDING POTENTIAL

Collective-intelligence architectural programming system for mega infrastructure projects 2023 - present

- Doctoral research project linking architectural programming theory, stakeholder coordination, and multi-agent reinforcement learning.
- Targets deployable decision-support workflows for early-stage programming of mega transportation projects.

Value management, stakeholder coordination, and complexity governance in megaprojects 2025 - present

- Visiting PhD research at The University of Melbourne, Department of Infrastructure Engineering.
- Connects infrastructure engineering management with architectural front-end decision-making and public-value evaluation.

Grant development themes for postdoctoral / early-career PI applications 2026 - 2031

- AI-enabled collaborative programming platforms for public buildings and transport hubs.
- Human-centered urban analytics for child-friendly communities, social infrastructure, and learning environments.
- Transparent megaproject governance models combining public sentiment, stakeholder networks, and risk simulation.

TEACHING, CURRICULUM DEVELOPMENT, AND MENTORING

The University of Melbourne, Melbourne, Australia Semester 2, 2025

Teaching Assistant, Engineering Management Capstone (ENGM90016).

- Supported graduate student teams in project formulation, business-case logic, proposal presentation, budgets, risk management, and project control.
- Teaching contribution emphasized decision criteria, economic and financial reasoning, communication, leadership, and collaborative engineering management.
- Course outcome reported on personal teaching page: ENGM90016 achieved Top 20% of ESS scores in EMI in Semester 2 of 2025.

Tongji University, Shanghai, China Spring 2024 - 2025

Teaching Assistant, AI Enabled Architectural Programming and Design.

- Guided graduate students in child-friendly public-space problem framing, social context analysis, field research, LLM/AIGC workflows, architectural programming, and evidence-based design deliverables.
- Contributed to course design linking public needs analysis, brief generation, design iteration, and multi-stakeholder evaluation.
- Developed and evaluated an AI-assisted architectural programming/design course model, published in *Buildings* in 2024.

Teaching and mentoring areas ongoing

- Architectural programming; AI for architectural design; construction/project management; research methods; evidence-based design and post-occupancy evaluation.
- Mentored student teams in capstone and design-studio settings from ambiguous problem contexts to structured briefs, defensible decisions, and final presentations.

ACADEMIC SERVICE AND PEER REVIEW

Verified peer reviewer, Web of Science / Publons 2024 - 2026

- 183 verified peer reviews across 11 journals, verified on Web of Science and ORCID.
- Review venues: *Engineering, Construction and Architectural Management* (89), *Buildings* (42), *Sustainability* (34), *SAGE Open* (7), *Geocarto International* (3), *Systems* (3), *Computer Applications in Engineering Education* (1), *Journal of Asian Architecture and Building Engineering* (1), *Journal of Construction Engineering and Management* (1), *Journal of Safety Research* (1), and *PLOS One* (1).
- WoS Core Collection author metrics: 11 indexed publications, WoS h-index 6, 90 times cited, 84 citing articles.

INVITED TALKS AND CONFERENCE PRESENTATIONS

31st International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA) May 2026

- Presented research on collective intelligence and multi-agent reinforcement learning for architectural programming in mega transportation projects, NYCU, Hsinchu, Taiwan, China.

60th ISOCARP World Planning Congress Oct. 2024

- Presented value-based architectural programming approach for enhancing children's well-being in inclusive urban community planning, Siena, Italy.

AWARDS, HONORS, AND COMPETITIVE FUNDING

- National Scholarship for PhD Students, Ministry of Education of the P.R.C., 2025.
- Outstanding Student, Tongji University, 2025.
- National Scholarship for PhD Students, Ministry of Education of the P.R.C., 2024.
- Outstanding Student, Tongji University, 2024.
- Outstanding Graduate, Huazhong University of Science and Technology, 2023.
- National Scholarship for Master Students, Ministry of Education of the P.R.C., 2022.
- Excellent Graduate Student, Huazhong University of Science and Technology, 2022.
- First-Class Scholarship, Huazhong University of Science and Technology, 2020, 2021, 2022, 2023.
- First Prize, National BIM Graduation Design Competition, 2020.
- Jilin Provincial Government Scholarship, Jilin Provincial Education Commission, 2019.
- First-Class Undergraduate Academic Scholarship, Jilin Jianzhu University.
- Second Prize, National College English Competition.

PROFESSIONAL PRACTICE

China Railway Siyuan Survey and Design Group, Wuhan, China Jul. 2022 - Aug. 2022

Architecture Management Intern.

- Contributed to construction drawings for Nantong Station and Changzhou South Station, and concept design for Wenzhou East Station urban transportation interchange center.

China Architecture Design & Research Group, Beijing, China Jul. 2021 - Sep. 2021

Architectural Designer Intern.

- Supported high-rise green building design, energy calculation, and construction drawings for Jinan Evergrande International Finance Center.

Wuhan Hechuang Architectural Design Co., Ltd., Wuhan, China Jun. 2019 - Aug. 2019

Architectural Designer Intern.

- Worked on campus planning and architectural design for Wuhan Optics Valley No. 13 Junior High School.

RESEARCH METHODS, TECHNICAL SKILLS, AND LANGUAGES

Computational

- Python
- Multi-agent reinforcement learning
- Machine learning and deep learning
- LLM fine-tuning and AIGC workflows
- Natural language processing

Analytical

- Architectural programming
- Topic and sentiment modeling
- Bibliometrics and systematic review
- SNA, MCDM, fuzzy/grey methods
- Monte Carlo simulation and POE

Design / Language

- Rhino, Grasshopper, Revit
- SketchUp, AutoCAD, ComfyUI
- SPSS, NVivo
- Mandarin Chinese: native
- English: fluent, IELTS 7.0

REFERENCES

Prof. Huijun Tu Tongji University

- Doctoral supervisor, College of Architecture and Urban Planning, Tongji University.

A/Prof Felix Kin Peng Hui The University of Melbourne

- Joint-training advisor, Department of Infrastructure Engineering, Faculty of Engineering and Information Technology; engineering technology management expert.

A/Prof Paulo Vaz Serra The University of Melbourne

- Joint-training advisor, Faculty of Architecture, Building and Planning; associate professor in construction management.